

Proofs Using Logical Equivalences

Rosen 1.2

List of Logical Equivalences

$$p \wedge T \Leftrightarrow p; \quad p \vee F \Leftrightarrow p$$

Identity Laws

$$p \vee T \Leftrightarrow T; \quad p \wedge F \Leftrightarrow F$$

Domination Laws

$$p \vee p \Leftrightarrow p; \quad p \wedge p \Leftrightarrow p$$

Idempotent Laws

$$\neg(\neg p) \Leftrightarrow p$$

Double Negation Law

$$p \vee q \Leftrightarrow q \vee p; \quad p \wedge q \Leftrightarrow q \wedge p$$

Commutative Laws

$$(p \vee q) \vee r \Leftrightarrow p \vee (q \vee r); \quad (p \wedge q) \wedge r \Leftrightarrow p \wedge (q \wedge r)$$

Associative Laws

List of Equivalences

$$p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$$

$$p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$$

$$\neg(p \vee q) \Leftrightarrow (\neg p \wedge \neg q)$$

$$\neg(p \wedge q) \Leftrightarrow (\neg p \vee \neg q)$$

$$p \vee \neg p \Leftrightarrow \mathbf{T}$$

$$p \wedge \neg p \Leftrightarrow \mathbf{F}$$

$$(p \rightarrow q) \Leftrightarrow (\neg p \vee q)$$

$$p \leftrightarrow q \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$$

Distribution Laws

De Morgan's Laws

Miscellaneous

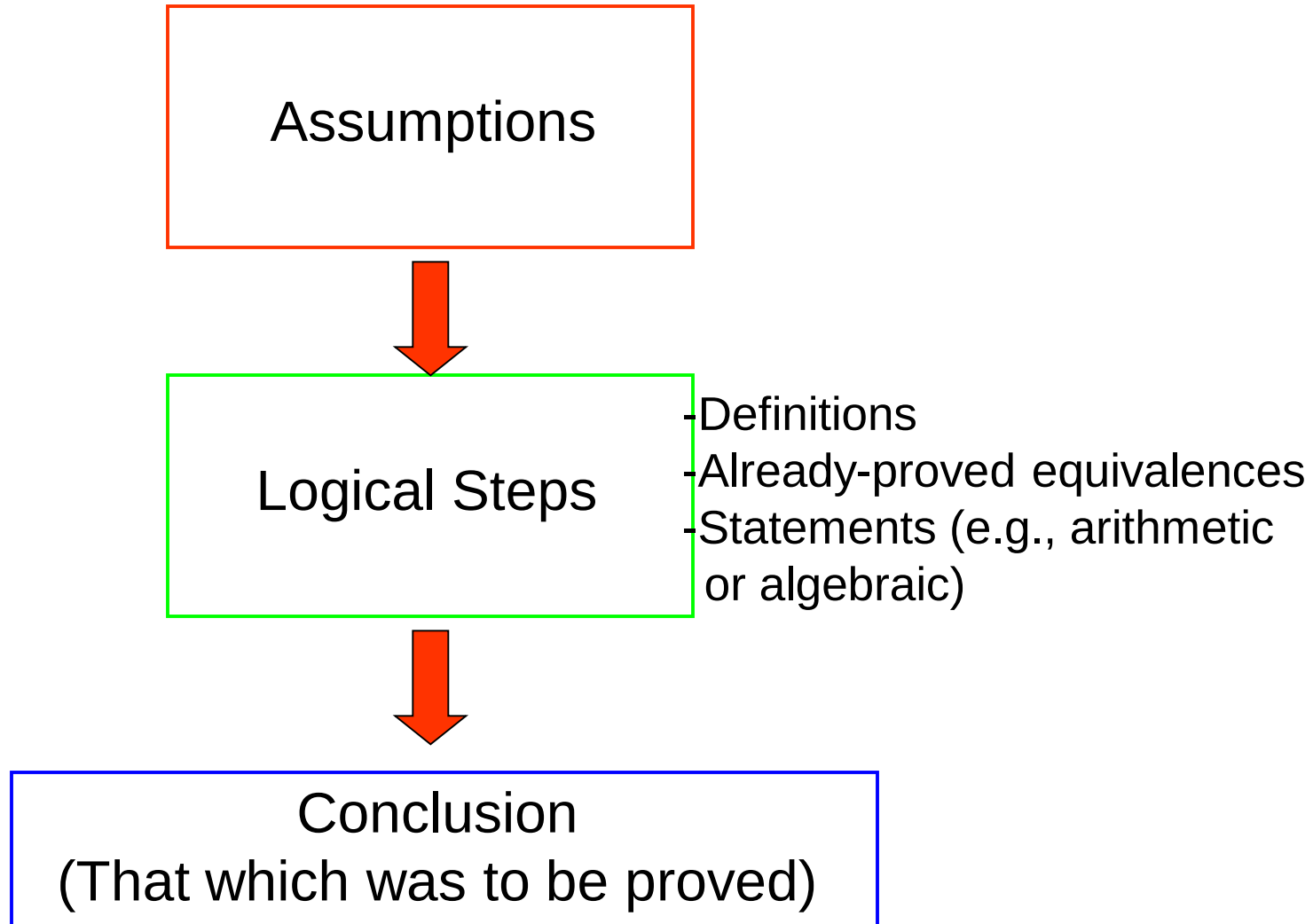
Or Tautology

And Contradiction

Implication Equivalence

Biconditional Equivalence

The Proof Process



Prove: $(p \wedge \neg q) \vee q \Leftrightarrow p \vee q$

$(p \wedge \neg q) \vee q$	Left-Hand Statement
$\Leftrightarrow q \vee (p \wedge \neg q)$	Commutative
$\Leftrightarrow (q \vee p) \wedge (q \vee \neg q)$	Distributive
$\Leftrightarrow (q \vee p) \wedge T$	Or Tautology
$\Leftrightarrow q \vee p$	Identity
$\Leftrightarrow p \vee q$	Commutative

Begin with exactly the left-hand side statement

End with exactly what is on the right

Justify EVERY step with a logical equivalence

Prove: $(p \wedge \neg q) \vee q \Leftrightarrow p \vee q$

$(p \wedge \neg q) \vee q$ Left-Hand Statement

$\Leftrightarrow q \vee (p \wedge \neg q)$ Commutative

$\Leftrightarrow (q \vee p) \wedge (q \vee \neg q)$ Distributive

Why did we need this step?



Our logical equivalence specified that $\vee$ is distributive on the right. This does not guarantee the equivalence works on the left!

Ex.: Matrix multiplication is not always commutative

(Note that whether or not $\vee$ is distributive on the left is not the point here.)

Prove: $p \rightarrow q \Leftrightarrow \neg q \rightarrow \neg p$

Contrapositive

$$p \rightarrow q$$

$$\Leftrightarrow \neg p \vee q \quad \text{Implication Equivalence}$$

$$\Leftrightarrow q \vee \neg p \quad \text{Commutative}$$

$$\Leftrightarrow \neg(\neg q) \vee \neg p \quad \text{Double Negation}$$

$$\Leftrightarrow \neg q \rightarrow \neg p \quad \text{Implication Equivalence}$$

Prove: $p \rightarrow p \vee q$ is a tautology

Must show that the statement is true for any value of p, q .

$$p \rightarrow p \vee q$$

$$\Leftrightarrow \neg p \vee (p \vee q)$$

Implication Equivalence

$$\Leftrightarrow (\neg p \vee p) \vee q$$

Associative

$$\Leftrightarrow (p \vee \neg p) \vee q$$

Commutative

$$\Leftrightarrow T \vee q$$

Or Tautology

$$\Leftrightarrow q \vee T$$

Commutative

$$\Leftrightarrow T$$

Domination

This tautology is called the addition rule of inference.

Why do I have to justify everything?

- Note that your operation must have the same order of operands as the rule you quote unless you have already proven (and cite the proof) that order is not important.
 - $3+4 = 4+3$
 - $3/4 \neq 4/3$
 - $A*B \neq B*A$ for everything (for example, matrix multiplication)

Prove: $(p \wedge q) \rightarrow p$ is a tautology

$$(p \wedge q) \rightarrow p$$

$$\Leftrightarrow \neg(p \wedge q) \vee p$$

Implication Equivalence

$$\Leftrightarrow (\neg p \vee \neg q) \vee p$$

DeMorgan's

$$\Leftrightarrow (\neg q \vee \neg p) \vee p$$

Commutative

$$\Leftrightarrow \neg q \vee (\neg p \vee p)$$

Associative

$$\Leftrightarrow \neg q \vee (p \vee \neg p)$$

Commutative

$$\Leftrightarrow \neg q \vee T$$

Or Tautology

$$\Leftrightarrow T$$

Domination

Prove or Disprove

$$p \rightarrow q \Leftrightarrow p \wedge \neg q ???$$

To prove that something is not true it is enough to provide one counter-example.
(Something that is true must be true in every case.)

p	q	$p \rightarrow q$	$p \wedge \neg q$
F	T	T	F

The statements are not logically equivalent

Prove: $\neg p \leftrightarrow q \Leftrightarrow p \leftrightarrow \neg q$

$$\neg p \leftrightarrow q$$

$$\Leftrightarrow (\neg p \rightarrow q) \wedge (q \rightarrow \neg p) \quad \text{Biconditional Equivalence}$$

$$\Leftrightarrow (\neg \neg p \vee q) \wedge (\neg q \vee \neg p) \quad \text{Implication Equivalence (x2)}$$

$$\Leftrightarrow (p \vee q) \wedge (\neg q \vee \neg p) \quad \text{Double Negation}$$

$$\Leftrightarrow (q \vee p) \wedge (\neg p \vee \neg q) \quad \text{Commutative}$$

$$\Leftrightarrow (\neg \neg q \vee p) \wedge (\neg p \vee \neg q) \quad \text{Double Negation}$$

$$\Leftrightarrow (\neg q \rightarrow p) \wedge (p \rightarrow \neg q) \quad \text{Implication Equivalence (x2)}$$

$$\Leftrightarrow p \leftrightarrow \neg q \quad \text{Biconditional Equivalence}$$